

PART-1 : MATHEMATICS

SECTION-I (i)

1) The sum $\sum_{k=0}^{49} (-1)^k {}^{99}C_{2k}$ equals

- (A) -2^{98}
- (B) 2^{98}
- (C) -2^{49}
- (D) 2^{49}

2) Let n be an even positive integer such that $n/2$ is odd and let $\alpha_0, \alpha_1, \alpha_2, \dots, \alpha_{n-1}$ be the complex

roots of unity of order n then the value of $\prod_{k=0}^{n-1} [(2+i) + (3-i)\alpha_k^2]$

- (A) $[(2+i)^{n/2} - (3-i)^{n/2}]^2$
- (B) $[(2+i)^{n/2} + (3-i)^{n/2}]^2$
- (C) $\left(\frac{2+i}{3-i}\right)^n$
- (D) $\left(\frac{2+i}{3-i}\right)^{n/2}$

3) $(x+m)^m - {}^mC_1(x+m-1)^m + {}^mC_2(x+m-2)^m \dots + (-1)^m x^m$ equals to :

- (A) $(m+1)!$
- (B) ${}^{m+1}C_m$
- (C) $m!$
- (D) ${}^mP_{m-2}$

4) Let z be a complex number such that $\left|2z + \frac{1}{z}\right| = 1$ and $\arg(z) = \theta$. Then minimum value of $\sin^2 \theta$.

- (A) $\frac{7}{8}$
- (B) $\frac{5}{4}$
- (C) $\frac{1}{5}$

(D) $\frac{1}{2}$

SECTION-I (ii)

1) Let z_1, z_2, z_3 and be three distinct complex numbers, satisfying $|z_1| = |z_2| = |z_3| = 1$, which of the following is/are correct.

(A) If $\arg\left(\frac{z_1}{z_2}\right) = \frac{\pi}{2}$ then $\arg\left(\frac{z-z_1}{z-z_2}\right) > \frac{\pi}{4}$ where $|z| > 1$

(B) $|z_1 z_2 + z_2 z_3 + z_3 z_1| = |z_1 + z_2 + z_3|$

(C) $\operatorname{Im}\left(\frac{(z_1 + z_2)(z_2 + z_3)(z_3 + z_1)}{z_1 \cdot z_2 \cdot z_3}\right) = 0$

If $|z_1 - z_2| = \sqrt{2}|z_1 - z_3| = \sqrt{2}|z_2 - z_3|$, then

(D) $\operatorname{Re}\left(\frac{z_3 - z_1}{z_3 - z_2}\right) = 0$

2) If z_1, z_2, z_3, z_4 are complex numbers in an Argand plane satisfying $z_1 + z_3 = z_2 + z_4$. A

complex number 'z' lies on the line joining z_1 and z_4 such that $\operatorname{Arg}\left(\frac{z-z_2}{z_1-z_2}\right) = \operatorname{Arg}\left(\frac{z_3-z_2}{z-z_2}\right)$. It is given that $|z - z_4| = 5$, $|z - z_2| = |z - z_3| = 6$ then

(A) area of the triangle formed by z, z_1, z_2 is $3\sqrt{7}$ sq. units

(B) area of triangle formed by z, z_3, z_4 is $\frac{15\sqrt{7}}{4}$ sq. units

(C) area of the quadrilateral formed by the points z_1, z_2, z_3, z_4 taken in order is $\frac{27\sqrt{7}}{2}$ sq. units

(D) area of the quadrilateral formed by the points z_1, z_2, z_3, z_4 taken in order is $\frac{47\sqrt{7}}{3}$ sq. units

3) If the complex numbers z_1, z_2, z_3 and z_4 , in order, are vertices of a quadrilateral such that $5z_1 - 6z_2 + 3z_3 - 2z_4 = 0$ and $5|z_1 - z_4|^2 = 9|z_2 - z_3|^2$ then which of the following is/are correct.

(A) z_1, z_2, z_3 and z_4 are vertices of a rectangle.

(B) The line segment joining z_1 and z_3 divides the line segment joining z_2 and z_4 in the ratio 1 : 3.

(C) The line segment joining z_2 and z_4 divides the line segment joining z_1 and z_3 in the ratio 3:5.

(D) z_1, z_2, z_3 and z_4 are concyclic.

SECTION-I (iii)

1)

Match the following List-I and List-II

	List-I		List-II
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(P)	The total number of three digit numbers, the sum of whose digits is even is equal to	(1)	18
(Q)	Total number of positive integral solutions of the equation $xyz = 140$ is equal to	(2)	54
(R)	Total number of positive integral solutions of $x + y + z \leq 10$ is equal to	(3)	120
(S)	If the cubic $x^3 + ax^2 + bx + c$ is divisible by $x^2 + 1$, then number of three digit numbers of the form abc or bca which can be formed is equal to	(4)	450
		(5)	150

(A) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 3$

(B) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 1$

(C) $P \rightarrow 4; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 2$

(D) $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 2$

2) Let $z_k = \cos\left(\frac{2k\pi}{10}\right) + i \sin\left(\frac{2k\pi}{10}\right); k = 1, 2, \dots, 9.$

	List-I		List-II
(P)	For each z_k there exists a z_j such that $z_k \cdot z_j = 1$	(1)	True
(Q)	There exists a $k \in \{1, 2, \dots, 9\}$ such that $z_1 \cdot z = z_k$ has no solution z in the set of complex numbers.	(2)	False
(R)	$\frac{ 1 - z_1 1 - z_2 \dots 1 - z_9 }{10}$ equals	(3)	1
(S)	$1 - \sum_{k=1}^9 \cos\left(\frac{2k\pi}{10}\right)$ equals	(4)	2
		(5)	4

(A) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 3$

(B) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 4$

(C) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 4$

(D) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 3$

3) Let z be a complex number satisfying $z^4 - 6z^3 + 18z^2 - 30z + 25 = 0$ has four roots z_1, z_2, z_3 and z_4 . Where $|z_1 - 1 - 2i| = 0$, $\operatorname{Re}(z_4) = \operatorname{Re}(z_1)$ and $\operatorname{Im}(z_2) > 0$ and $\bar{z}$ denotes the complex conjugate of z . Match each entry in List-I to the correct entries in List-II.

	List-I		List-II
(P)	The area of the quadrilateral bounded by z_1, z_2, z_3 & z_4 is	(1)	1
(Q)	$\frac{ (z_0 - \bar{z}_1)(z_0 - \bar{z}_3) }{ (z_0 - \bar{z}_4)(z_0 - \bar{z}_2) }$ is equal to (where z^0 is any purely real no.)	(2)	2

(R)	If $(\alpha - z_1 + \alpha - z_2 + \alpha - z_3 + \alpha - z_4)$ is minimum then $3 \alpha $ is (where α is a complex number)	(3)	3
(S)	If $\arg \left(\frac{\beta - z_1}{\beta - z_3} \right) = \tan^{-1} \left(\frac{2}{1 + \sqrt{5}} \right) + \tan^{-1} \left(\frac{1}{2 + \sqrt{5}} \right)$ then $\frac{1}{\sqrt{5}} \beta - 2 - i _{\max}$ is	(4)	4
		(5)	5

Then the correct option is

- (A) $P \rightarrow 3; Q \rightarrow 3; R \rightarrow 4; S \rightarrow 3$
 (B) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 4$
 (C) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 5; S \rightarrow 2$
 (D) $P \rightarrow 1; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 5$

4) Let
$$P(n+1) = \sum_{k=0}^n \frac{(-1)^{n-k} \cdot k}{k+1} \frac{(n+1)!}{k!(n+1-k)!}$$

List-I		List-II	
(P)	P(8)	(1)	$\frac{3}{4}$
(Q)	P(7)	(2)	$\frac{4}{5}$
(R)	P(9)	(3)	1
(S)	P(11)	(4)	$\frac{5}{6}$
		(5)	$\frac{1}{4}$

- (A) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 5$
 (B) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 4$
 (C) $P \rightarrow 5; Q \rightarrow 3; R \rightarrow 4; S \rightarrow 1$
 (D) $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 5; S \rightarrow 3$

SECTION-II

1) Let $(1+x)^{1/x} = a_0 + a_1x + a_2x^2 + a_3x^3 + \dots \infty$ If $a_2 = \frac{m}{n} \cdot e$ and $a_3 = \frac{p}{q} \cdot e$ where $m, p \in \mathbb{I}$ and $n, q \in \mathbb{N}$ such that H.C.F. of $|m|$ & n as well as $|p|$ and q are 1 each. Then evaluate $|m| + |p| + n + q$.

2) Let
$$a = \sum_{n=0}^{\infty} \frac{x^{3n}}{(3n)!}, \quad b = \sum_{n=0}^{\infty} \frac{x^{3n+1}}{(3n+1)!} \quad \text{and} \quad c = \sum_{n=0}^{\infty} \frac{x^{3n+2}}{(3n+2)!}$$
 Evaluate $a^3 + b^3 + c^3 - 3abc$.

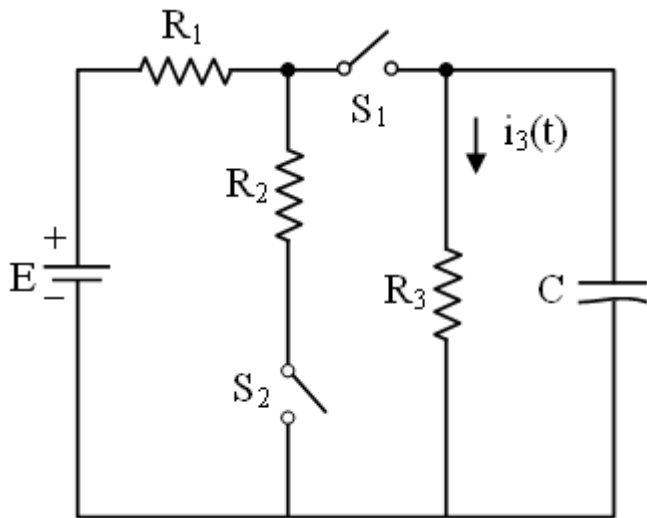
- 3) Value of the remainder when $\sum_{r=0}^{2014} \sum_{k=0}^r (-1)^k (k+1)(k+2) {}^{2019}C_{r-k}$ is divided by 64
- 4) The number of five digit numbers which are divisible by 3 that can be formed by using the digits 1,2,3,4,5,6,7,8,9 when repetition of digits is allowed, are 3^k , then k is equal to
- 5) Find the number of ways in which two Americans, two British, one Chinese, one Dutch and one Egyptian can sit on a round table so that person of the same nationality are separated ?

- 6) If m and x are two real numbers, then $e^{2mi \cot^{-1} x} \left(\frac{xi+1}{xi-1} \right)^m$ is equal to

PART-2 : PHYSICS

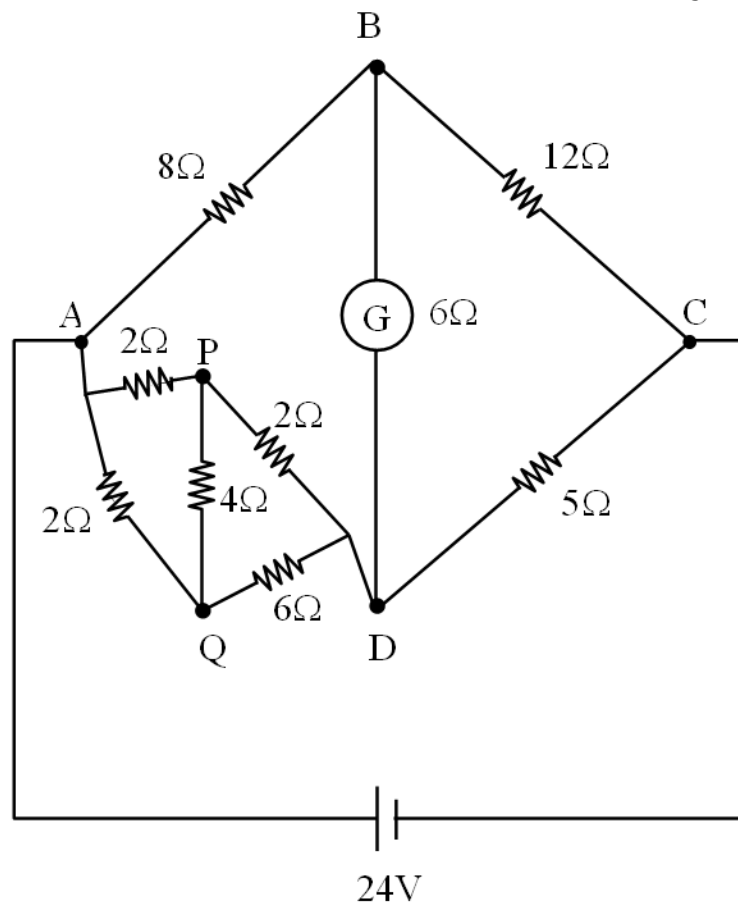
SECTION-I (i)

- 1) A uniform wire made of a monovalent metal has resistance 1.000Ω at 100 K and 2.0808Ω at 300 K . The metal expands isotropically, with a temperature-independent linear expansion coefficient $\alpha = 1.0 \times 10^{-4} \text{ K}^{-1}$. Assume that the number of conduction electrons in the wire remains constant and that the effective electron mass is independent of temperature. According to the Drude model, the ratio of relaxation times is $\frac{\tau_{100}}{\tau_{300}} =$
- (A) 1.92
(B) 2.00
(C) 2.04
(D) 2.08
- 2) In the circuit shown, the capacitor is initially uncharged. At $t = 0$, switch S_1 is closed while S_2 remains open. At $t = T = 0.40 \text{ s}$, switch S_2 is also closed and S_1 remains closed. Which of the following expressions correctly gives the current $i_3(t)$ through resistor R_3 for $t \geq 0.40 \text{ s}$? [Given : $E = 24 \text{ V}$, $R_1 = 60 \text{ k}\Omega$, $R_2 = 40 \text{ k}\Omega$, $R_3 = 120 \text{ k}\Omega$, $C = 10 \mu\text{F}$, $T = 0.40 \text{ s}$]



- (A) $i_3(t) = [0.0667 + 0.0176 e^{-5(t - 0.40)}] \text{mA}$
 (B) $i_3(t) = [0.0667 - 0.0176 e^{-5(t - 0.40)}] \text{mA}$
 (C) $i_3(t) = 0.0843 e^{-5(t - 0.40)} \text{mA}$
 (D) $i_3(t) = [0.1333 - 0.0667 e^{-2.5t}] \text{mA}$

3) In the Wheatstone bridge ABCD, an ideal 24 V battery is connected between A and C, and a galvanometer of resistance 6Ω is connected between B and D. The outer arms are $AB = 8 \Omega$, $BC = 12 \Omega$, $CD = 5 \Omega$, while the arm AD is itself a Wheatstone bridge. The inner bridge between A and D has $AP = 2 \Omega$, $PQ = 4 \Omega$, $QD = 6 \Omega$, and $AQ = 2 \Omega$, $PD = 2 \Omega$. The current through the outer

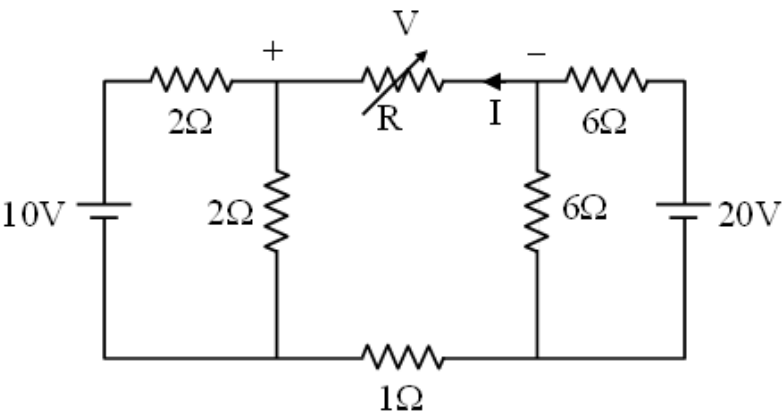


galvanometer is closest to :

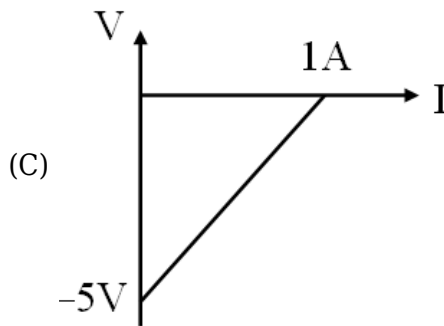
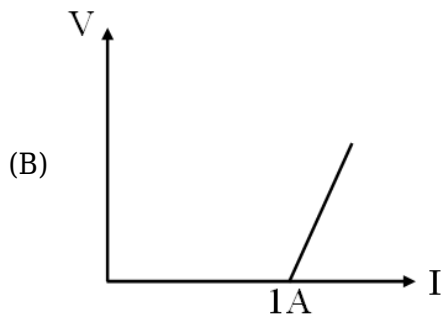
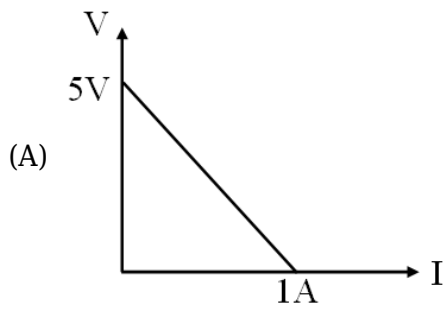
- (A) 0.11 A from B to D

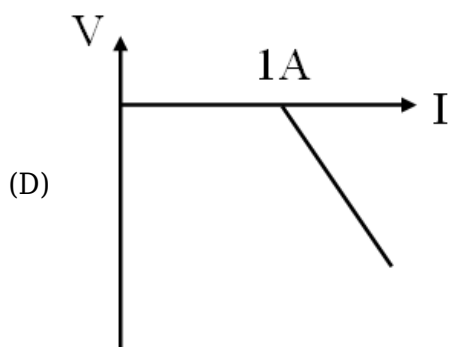
- (B) 0.32 A from B to D
- (C) 0.11 A from D to B
- (D) 0.32 A from D to B

4) Consider circuit as shown. Which of the graph best represent V-I relationship across the variable



resistance R.

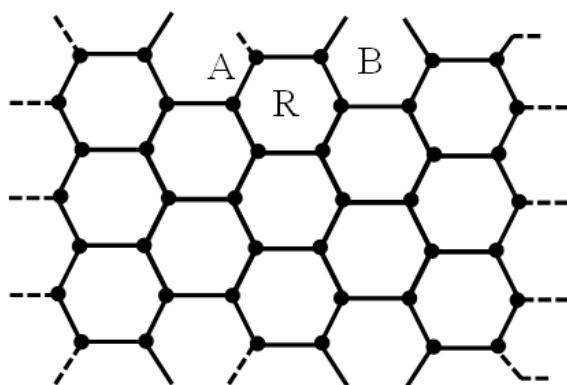




SECTION-I (ii)

1) Two identical infinite honeycomb resistor meshes lie in two parallel planes Π_1 and Π_2 . Every edge of each honeycomb contains a resistor of resistance R . The two meshes are perfectly aligned so that every vertex of Π_1 lies directly above a corresponding vertex of Π_2 . Each pair of corresponding vertices is connected by a resistor of resistance λR , $\lambda > 0$. Choose two adjacent vertices A and B in the upper mesh. Let A' and B' be the corresponding vertices directly below them in the lower mesh. The vertices A and A' are shorted by an ideal wire to form terminal P , while B and B' are shorted by another ideal wire to form terminal Q : $A \equiv A' \equiv P$, $B \equiv B' \equiv Q$. An ideal battery of emf V is connected between P and Q . Apart from these two external shorting wires, corresponding vertices of the two planes are connected only through their respective λR resistors. Both honeycomb lattices extend infinitely in all directions in their respective planes. Which of the following statements are

Top view (upper plane π_1)



correct ?

(A) The complete two-layer network draws the same current from the battery as a single resistor of resistance $\frac{R}{3}$ and this equivalent resistance is independent of λ .

(B) Every inter-plane resistor of resistance λR carries zero current. Further, the sum of the currents through the two directly corresponding edges AB and $A'B'$ is $\frac{V}{R}$.

(C) Suppose only the resistors AB and $A'B'$ are removed, while all other resistors of both infinite meshes and all inter-plane resistors λR remain unchanged. The equivalent resistance between P and Q then becomes R .

(D) Since the inter-plane resistors carry zero current in the above arrangement, the entire double-layer network is equivalent to a single honeycomb mesh having resistance $R/2$ on each edge for any arbitrary choice of two terminals in the double-layer network.

2) An analog multimeter uses a moving-coil galvanometer of resistance $G = 100\Omega$ and full-scale

current $I_g = 1.00 \text{ mA}$. In ammeter mode, suitable shunts are connected across the galvanometer to obtain 10 mA and 100 mA ranges. In voltmeter mode, suitable series resistors are used to obtain 10V and 50 V ranges. In resistance mode, the galvanometer is connected in series with an ideal 1.50 V cell and an adjustable resistance. With the test terminals shorted, the adjustable resistance is set so that the galvanometer gives full-scale deflection. Which of the following statements are correct ?

- (A) The required shunts for the 10mA and 100mA ranges are, respectively, $\frac{100}{9} \Omega$ and $\frac{100}{99} \Omega$, while the series resistances required for the 10 V and 50 V ranges are 4.95 k Ω and 24.95 k Ω , respectively.
- On the 10 V range, the voltmeter has an input resistance of 10k Ω . If it is connected across a 20
- (B) k Ω resistor which is in series with a 10k Ω resistor across an ideal 12 V source, the voltmeter reads 4.8V.
- In resistance mode, after zero adjustment, the total internal series resistance of the multimeter
- (C) is 1.50 k Ω . An external resistance of 1.50 k Ω therefore gives half-scale deflection, whereas 4.50 k Ω gives one-fourth of full-scale deflection.
- (D) On both the 10mA and 100mA ranges, the effective resistance of the ammeter is the same because the galvanometer reaches full scale at the same terminal voltage of 0.10 V.

3) A cube ABCDEFGH is formed using capacitors along its 12 edges. The faces ABCD and EFGH are parallel, with AE, BF, CG, DH as the remaining four edges.

A capacitor of capacitance C is placed on every edge except BC, where the capacitance is kC, with $k > 0$. A battery of potential difference V is connected between the diagonally opposite vertices A and G. All other vertices are isolated and initially uncharged. Which of the following statements are correct?

- (A) The equivalent capacitance between A and G is $C_{eq} = \frac{2(7k+5)}{11k+9} C$.
- As k is increased continuously from 0 to ∞ , ($k = 0$ implying absence of capacitor BC) the
- (B) equivalent capacitance increases monotonically from $\frac{10C}{9}$ to $\frac{14C}{11}$.
- (C) The magnitude of charge on the capacitor BC is $Q_{BC} = \frac{4k}{11k+9} CV$.
- (D) In the limit $k \rightarrow \infty$, the potential difference across BC tends to zero; hence the charge stored on the capacitor BC also tends to zero.

SECTION-I (iii)

1)

	List-I (Circuits)		List-II (Time constant τ)
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(P)		(1)	20.4 ms
(Q)		(2)	24 ms
(R)		(3)	25.2 ms
(S)		(4)	30.8 ms
		(5)	41.2 ms

(A) P → 2; Q → 4; R → 3; S → 5

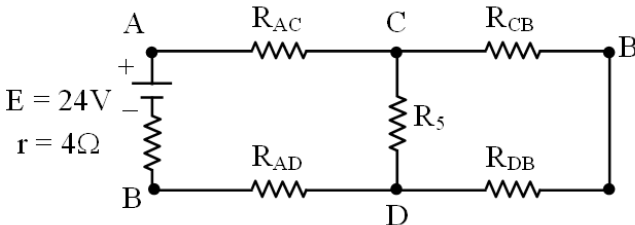
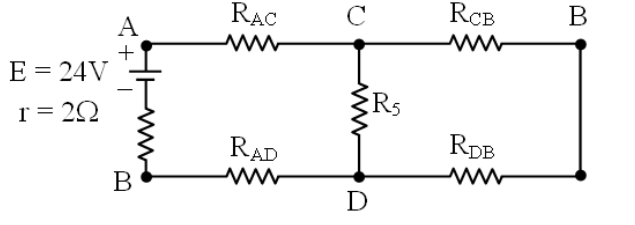
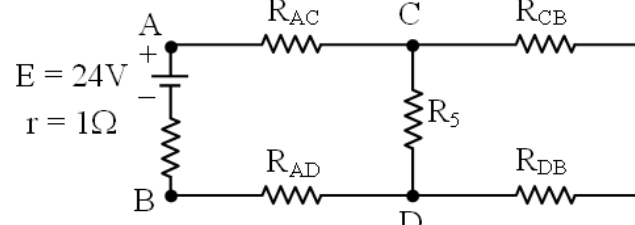
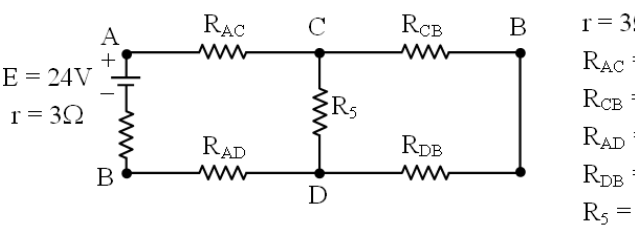
(B) P → 4; Q → 2; R → 1; S → 3

(C) P → 3; Q → 4; R → 2; S → 5

(D) P → 2; Q → 4; R → 3; S → 1

2)

List-I (Circuits)	List-II (Power dissipated in R_5)
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(P)	 $E = 24\text{V}$ $r = 4\Omega$ $r = 4\Omega$ $R_{AC} = 1\Omega$ $R_{CB} = 4\Omega$ $R_{AD} = 5\Omega$ $R_{DB} = 6\Omega$ $R_5 = 2\Omega$	(1)	0.50 W
(Q)	 $E = 24\text{V}$ $r = 2\Omega$ $r = 2\Omega$ $R_{AC} = 1\Omega$ $R_{CB} = 3\Omega$ $R_{AD} = 7\Omega$ $R_{DB} = 5\Omega$ $R_5 = 1\Omega$	(2)	1.00 W
(R)	 $E = 24\text{V}$ $r = 1\Omega$ $r = 1\Omega$ $R_{AC} = 1\Omega$ $R_{CB} = 7\Omega$ $R_{AD} = 3\Omega$ $R_{DB} = 5\Omega$ $R_5 = 2\Omega$	(3)	2.00 W
(S)	 $E = 24\text{V}$ $r = 3\Omega$ $r = 3\Omega$ $R_{AC} = 1\Omega$ $R_{CB} = 5\Omega$ $R_{AD} = 7\Omega$ $R_{DB} = 2\Omega$ $R_5 = 1\Omega$	(4)	4.00 W
		(5)	8.00 W

(A) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 5; S \rightarrow 3$

(B) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 4$

(C) $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 3; S \rightarrow 4$

(D) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 5$

3) Six large conducting plates $P_1, P_2, \dots, P_6$, each of area A , are arranged parallel to one another in this order with equal separation d .

The dielectric constants in the five successive gaps are $k_1 = 1$, $k_2 = 2$, $k_3 = 3$, $k_4 = 2$, $k_5 = 1$

so that, defining $C^0 = \frac{\epsilon_0 A}{d}$, the five adjacent-gap capacitances are $C^0, 2C^0, 3C^0, 2C^0, C^0$ respectively.

A : connected to terminal A B : connected to terminal B, F : isolated neutral plate X : interconnected and isolated conductor (net charge zero)

	List-I : Physical arrangements (gaps shown with capacitances)		List-II Equivalent capacitance between A and B
(P)		(1)	$\frac{39}{11}C_0$
(Q)		(2)	$\frac{17}{11}C_0$
(R)		(3)	$\frac{5}{2}C_0$
(S)		(4)	$\frac{20}{9}C_0$
		(5)	$\frac{15}{8}C_0$

(A) $P \rightarrow 2; Q \rightarrow 5; R \rightarrow 4; S \rightarrow 1$

(B) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 1$

(C) $P \rightarrow 2; Q \rightarrow 5; R \rightarrow 3; S \rightarrow 4$

(D) $P \rightarrow 4; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 5$

4) Each capacitor in List-I carries charges $+Q$ and $-Q$ on its two conductors. All dielectric materials are linear and isotropic. Neglect fringing and edge effects. Match each arrangement in List-I with the corresponding electrostatic energy in List-II.

	List-I		List-II
(P)	A capacitor is formed by two rectangular conducting plates of dimensions $L \times b$. The local separation varies as $d(x) = d\left(1 + \frac{x}{L}\right)$, $0 \leq x \leq L$. Assume $d \ll L$ and $d \ll b$, so that the local parallel-plate approximation is valid. The space between the plates is vacuum. Let $A = Lb$.	(1)	$\frac{Q^2}{24\pi\epsilon_0 R}$
(Q)	A spherical capacitor consists of concentric conducting spheres of radii R and $2R$. The region $R < r < \frac{3R}{2}$ is filled with a dielectric of relative permittivity 2, while $\frac{3R}{2} < r < 2R$ is vacuum.	(2)	$\frac{Q^2 \theta}{2\epsilon_0 L}$
(R)	A long cylindrical capacitor consists of coaxial conducting cylinders of radii a and $2a$ and length L , where $L \gg a$. The space between them contains a dielectric whose relative permittivity varies as $K(r) = \frac{r}{a}$.	(3)	$\frac{Q^2 d}{2\epsilon_0 A \ln 2}$
(S)	Two conducting radial planes form a wedge of angle θ (in radians). Their common length perpendicular to the cross-section is L , and they extend from $r = a$ to $r = 2a$. The region between them contains a dielectric with $K(r) = \frac{r}{a}$.	(4)	$\frac{Q^2}{8\pi\epsilon_0 L}$
		(5)	$\frac{Q^2 d \ln 2}{2\epsilon_0 A}$

(A) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 2$

(B) $P \rightarrow 5; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 2$

(C) $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 1; S \rightarrow 2$

(D) $P \rightarrow 5; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 1$

SECTION-II

1) A metre bridge has small but appreciable end corrections. If the balance point is at a distance

ℓ cm from the left end of the bridge wire, the balance condition is $\frac{P}{Q} = \frac{\ell + \alpha}{100 - \ell + \beta}$, where P and Q are the resistances in the left and right gaps respectively, while α and β are the unknown end corrections. Three observations are made :

Observation I:

A 2Ω resistance is placed in the left gap and a 5Ω resistance in the right gap. The balance point is obtained at $\ell = 28$ cm.

Observation II:

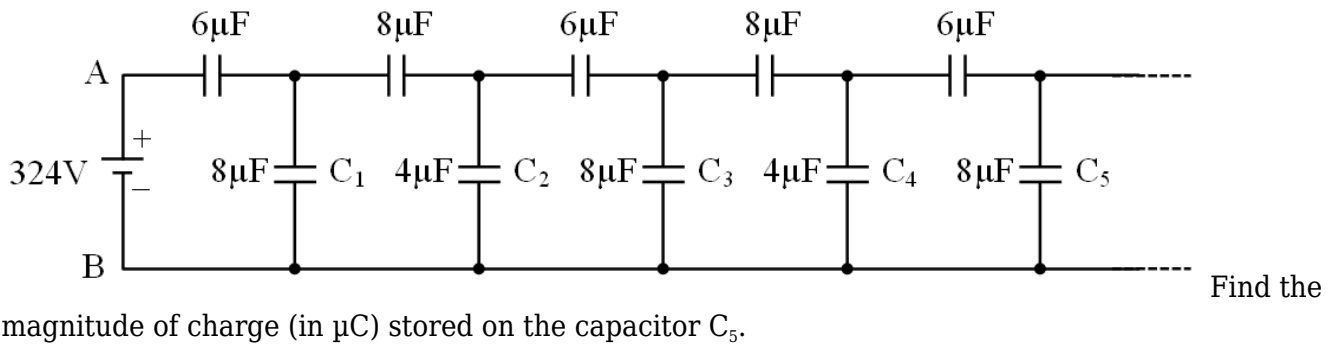
An unknown resistance X is placed in the left gap and a 6Ω resistance in the right gap. The balance point is obtained at $\ell = 40$ cm.

Observation III:

The positions of X and 6Ω are interchanged. The new balance point is $\ell = 61$ cm.

Assume that the bridge wire is otherwise uniform and the jockey contact resistance is negligible. Find X , in ohms.

2) An infinite ladder network of capacitors is connected across an ideal DC source of 324 V between terminals A and B as shown in the figure. Along the top branch, the capacitors have capacitances $6\ \mu\text{F}$, $8\ \mu\text{F}$, $6\ \mu\text{F}$, $8\ \mu\text{F}$, $6\ \mu\text{F}$, ... in this repeating order from left to right. From each top node, a capacitor is connected vertically to the common bottom rail B with capacitances $8\ \mu\text{F}$, $4\ \mu\text{F}$, $8\ \mu\text{F}$, $4\ \mu\text{F}$, $8\ \mu\text{F}$, ... in the same repeating order from left to right. Initially, all capacitors are uncharged. The circuit is allowed to reach electrostatic equilibrium.



3) A parallel-plate capacitor has rectangular plates of length $L = 40\text{ cm}$, width $w = 32\text{ cm}$, and separation $d = 3.0\text{ mm}$. A dielectric slab of dielectric constant $K = 4$, thickness $t = 2.0\text{ mm}$, and width equal to that of the plates can slide horizontally between the plates. The slab remains in contact with the lower plate, so that in the portion occupied by the slab there is an air layer of thickness 1.0 mm between the slab and the upper plate. Initially, the slab is inserted through a distance $x_0 = 10\text{ cm}$. With the slab in this position, the capacitor is connected to an ideal 6000 V battery and allowed to reach electrostatic equilibrium. The battery is then disconnected without changing the position of the slab. The slab is subsequently moved quasistatically further into the capacitor. When its insertion depth becomes $x = 20\text{ cm}$. Find the magnitude of the electrostatic force pulling the slab further into the capacitor. Neglect fringing effects and take $\epsilon_0 = 9 \times 10^{-12}\text{ Fm}^{-1}$. Give the answer in mN.

4) A parallel-plate capacitor has plate area $A = 200\text{ cm}^2$ and separation $d = 1.0\text{ mm}$. The region between the plates is divided into two equal-area parts, each extending through the full plate separation. The two regions contain weakly conducting dielectrics D_1 and D_2 :

$$K_1 = 3, \sigma_1 = 12\text{ pSm}^{-1}$$

$$K_2 = 5, \sigma_2 = \text{unknown.}$$

Each dielectric may be treated as an ideal capacitor in parallel with its corresponding leakage resistance. The capacitor is connected across an ideal 400 V battery for a sufficiently long time. The battery is then disconnected and, at the same instant, an initially uncharged ideal capacitor of capacitance $C_0 = 720\text{ pF}$ is connected in parallel with the leaky capacitor. The capacitor C_0 has no leakage. After 8s, the common potential difference across the combination is measured to be $\frac{200}{e^2}\text{ V}$. Neglect fringing effects and take $\epsilon_0 = 9 \times 10^{-12}\text{ Fm}^{-1}$. Find σ_2 in pS m^{-1} .

5) Two nonlinear two-terminal elements X and Y are connected in parallel. Their static I-V characteristics for $V \geq 0$ are

$$I_X = \frac{V^2}{8} A, I_Y = \frac{V^3}{32} A$$

where V is the voltage across each element. This parallel combination is connected in series with a 2Ω resistor across an ideal DC voltage source of voltage V_s . Initially, $V_s = 12V$, and the circuit reaches a steady operating point. The source voltage V_s is then varied by an infinitesimally small amount about $12V$.

If I denotes the current supplied by the source, determine the differential resistance

$$r_d = \left. \frac{dV_s}{dI} \right|_{V_s=12V}$$

Enter the value of $10^3 r_d$, where r_d is measured in ohms.

6) A potentiometer is used to determine the internal resistance of a cell X. The potential gradient of the potentiometer is not the same in different observations because the rheostat in the primary circuit is adjusted between measurements. A standard cell of emf $1.50V$ is used to determine the potential gradient each time.

Observation I : The standard cell balances at 300 cm . Under the same setting, cell X, on open circuit, balances at 240 cm .

Observation II : The rheostat setting is changed. The standard cell now balances at 375 cm . Cell X is connected across an unknown resistance R , and its terminal voltage balances at 225 cm .

Observation III : The rheostat setting is again changed. The standard cell now balances at 350 cm . A 6Ω resistor is connected in parallel with R , while cell X remains connected across this combination. The balance point for the terminal voltage of X is now 168 cm .

Assume that the potentiometer draws no current from the secondary circuit at balance and all connecting wires have negligible resistance. Find the internal resistance r of cell X, in ohms.

PART-3 : CHEMISTRY

SECTION-I (i)

1) The complementary DNA strand for the following strand of DNA is

$5' \leftarrow A - T - G - C - T \rightarrow 3'$

(A) $5' \leftarrow T - A - C - G - A \rightarrow 3'$

(B) $3' \leftarrow G - C - A - T - C \rightarrow 5'$

(C) $3' \leftarrow T - A - C - G - A \rightarrow 5'$

(D) $5' \leftarrow G - C - A - T - C \rightarrow 3'$

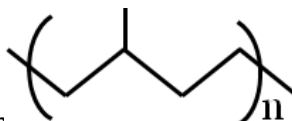
2) Which of the following statements is NOT correct about thermoplastic polymers?

(A) These are softened on heating.

(B) Intermolecular forces of attraction in these are in between elastomers and fibres.

(C) These possess extensive cross linking by covalent bonds.

(D) These are easily moulded.



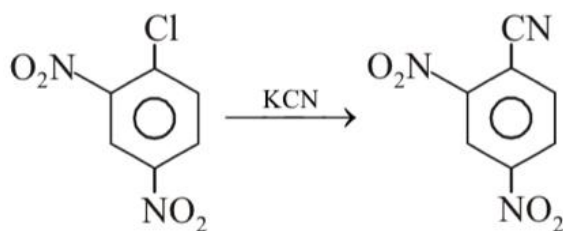
3) Select the correct statement about the polymer

- (A) It can be seen as addition homopolymer of isoprene
- (B) It can be seen as addition copolymer of ethylene & propylene
- (C) It can be vulcanized using sulfur to enhance cross-links
- (D) It possible monomers give -ve test with bromine water as well as with Baeyer's reagent

4) Which of the following amino acids moves towards cathode on electrophoresis at pH = 7 ?

- (A) Leucine
- (B) Lysine
- (C) Asparagine
- (D) Aspartic acid

SECTION-I (ii)



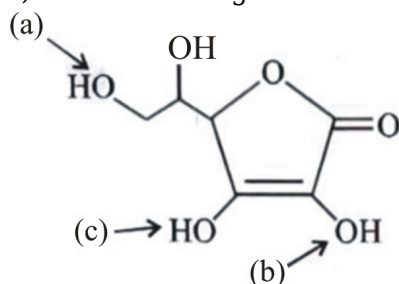
1) Select the correct statements about the above reaction.

- (A) The mechanism of the reaction is S_N2Ar .
- (B) Carbanion intermediate is formed in the reaction.
- (C) The rate of reaction is independent of the concentration of cyanide ion.
- (D) The rate of reaction decreases if Cl is replaced by F.

2) Which of the following statements are correct?

- (A) Cationic detergents have germicidal properties
- (B) Bacteria can degrade the detergents containing highly branched chains.
- (C) Some synthetic detergents can give foam even in ice cold water.
- (D) Synthetic detergents are not soaps.

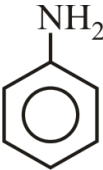
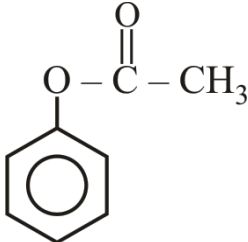
3) The structure given below is of ascorbic acid. Select the correct statements-



- (A) The hydrogen labelled (b) is most acidic.
 (B) It is water soluble and can be stored in body
 (C) It is found in citrus fruits
 (D) Its deficiency causes pernicious anaemia

SECTION-I (iii)

1)

List-I		List-II	
(P)		(1)	Most reactive for aromatic electrophile substitution reaction in neutral medium.
(Q)		(2)	It will produce salicylaldehyde when treated with $\text{CHCl}_3/\text{NaOH}$
(R)		(3)	It will produce tribromo derivative with $\text{Br}_2 / \text{H}_2\text{O}$
(S)		(4)	It will produce carboxylic acid on hydrolysis
		(5)	It will produce an explosive substance when treated with nitric acid in excess.

The correct combination is :

- (A) $\text{P} \rightarrow 2; \text{Q} \rightarrow 4; \text{R} \rightarrow 1; \text{S} \rightarrow 5$
 (B) $\text{P} \rightarrow 1; \text{Q} \rightarrow 4; \text{R} \rightarrow 3; \text{S} \rightarrow 5$
 (C) $\text{P} \rightarrow 2; \text{Q} \rightarrow 1; \text{R} \rightarrow 4; \text{S} \rightarrow 3$
 (D) $\text{P} \rightarrow 2; \text{Q} \rightarrow 5; \text{R} \rightarrow 3; \text{S} \rightarrow 4$

2) Match the list-

List-I (Reaction)		List-II (Observation)	
(P)	Aniline + CHCl_3 , NaOH, 70°C	(1)	Formation of salt that is soluble in alkali
(Q)	Aniline + NaNO_2/HCl , $0-5^\circ\text{C}$, then β -Naphthol	(2)	E_2 , alkene formed on larger ring

(R)	Isopropyl amine + PhSO_2Cl , then KOH	(3)	Reaction involves neutral divalent carbon intermediate (CCl_2)
(S)	1-Bromo bicyclo(2.2.1)heptane, alc. KOH , Δ	(4)	Brilliant orange - red dye is formed
		(5)	No reaction

(A) $\text{P} \rightarrow 3; \text{Q} \rightarrow 4; \text{R} \rightarrow 1; \text{S} \rightarrow 2$

(B) $\text{P} \rightarrow 5; \text{Q} \rightarrow 2; \text{R} \rightarrow 3; \text{S} \rightarrow 4$

(C) $\text{P} \rightarrow 2; \text{Q} \rightarrow 5; \text{R} \rightarrow 4; \text{S} \rightarrow 1$

(D) $\text{P} \rightarrow 3; \text{Q} \rightarrow 4; \text{R} \rightarrow 1; \text{S} \rightarrow 5$

3) Match the list

	List - I (Compounds)		List - II (Tests/Properties)
(P)	One of the monomers of Bakelite	(1)	Positive test with neutral FeCl_3 solution
(Q)	Tyrosine	(2)	Red precipitate of Cu_2O with Fehling solution
(R)	Fructose	(3)	Decolorizes Baeyer's reagent
(S)	One of the monomers of Buna-S	(4)	Can form osazone with excess PhNHNH_2
		(5)	Gives positive Biuret test

(A) $\text{P} \rightarrow 1,5; \text{Q} \rightarrow 2,5; \text{R} \rightarrow 3,4; \text{S} \rightarrow 1,3$

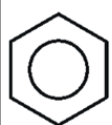
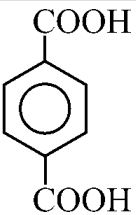
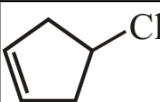
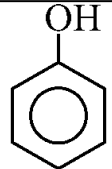
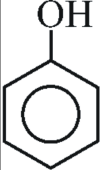
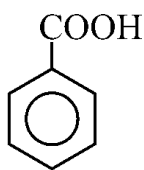
(B) $\text{P} \rightarrow 1,2; \text{Q} \rightarrow 1; \text{R} \rightarrow 2,4; \text{S} \rightarrow 3$

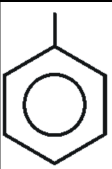
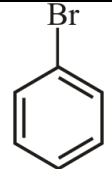
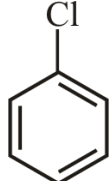
(C) $\text{P} \rightarrow 1,4; \text{Q} \rightarrow 1,5; \text{R} \rightarrow 2,5; \text{S} \rightarrow 1,5$

(D) $\text{P} \rightarrow 1,5; \text{Q} \rightarrow 3,4; \text{R} \rightarrow 2,4; \text{S} \rightarrow 1,3$

4)

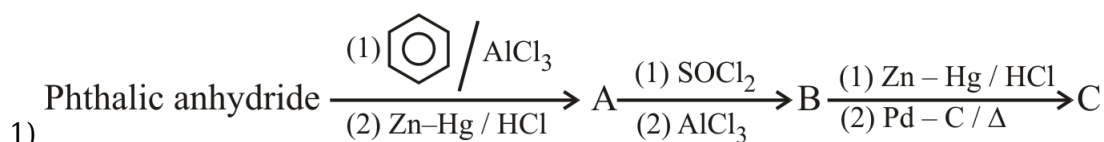
Match the list-

	List-I		List-II
(P)	 $\xrightarrow[\text{(iii) } \text{H}^+ / \text{H}_2\text{O}]{\text{(i) } \text{CH}_3\text{CHClCH}_3 / \text{Anhy. AlCl}_3, \text{(ii) } \text{O}_2 / h\nu}$	(1)	
(Q)	 $\xrightarrow[\text{(iii) } \text{CHCl}_2\text{Br} / \text{KOH}]{\text{(i) aq. KOH, (ii) } \text{H}^+ / \Delta}$	(2)	
(R)	 $\xrightarrow[\text{(iii) Zn dust, } \Delta]{\text{(i) } \text{CCl}_4 / \text{NaOH, (ii) } \text{H}^+}$	(3)	

(S)	 $\xrightarrow[\text{(ii) KMnO}_4 / \text{H}^+ / \Delta]{\text{(i) CH}_3\text{-C(=O)-Cl / Anhy. AlCl}_3}$	(4)	
		(5)	

- (A) P $\rightarrow$ 1; Q $\rightarrow$ 2; R $\rightarrow$ 3; S $\rightarrow$ 4
 (B) P $\rightarrow$ 2; Q $\rightarrow$ 4; R $\rightarrow$ 3; S $\rightarrow$ 1
 (C) P $\rightarrow$ 2; Q $\rightarrow$ 5; R $\rightarrow$ 3; S $\rightarrow$ 1
 (D) P $\rightarrow$ 2; Q $\rightarrow$ 4; R $\rightarrow$ 5; S $\rightarrow$ 3

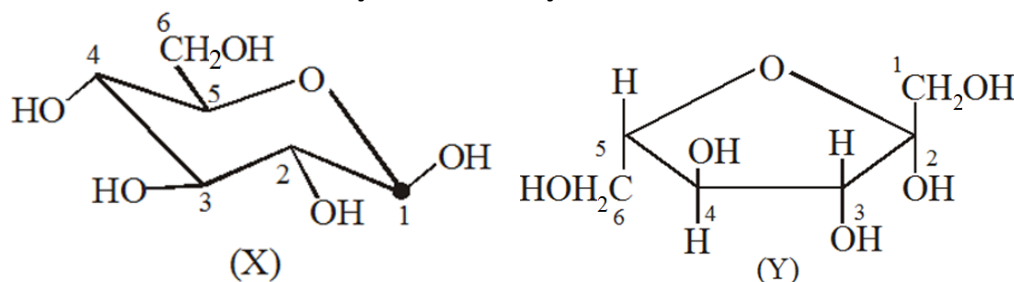
SECTION-II



Calculate the sum of degree of unsaturation of major organic products A, B and C. Assume acidic/basic workup, if necessary.

2) Nylon-610 is a copolymer of 2 different monomers. The monomer having lower molecular mass is subjected to Dumas method. Calculate the moles of nitrogen gas evolved from two moles of the monomer.

3) A non-reducing disaccharide is obtained by condensation of two monosaccharides X & Y given below. The condensation takes place between C-(x) of X and C-(y) of Y. Calculate the value of x + y (where, x & y are carbon atoms labelled in the diagram)



4) The structure of sucralose (artificial sweetener) is similar to that of sucrose, with three -OH groups replaced by -Cl & different configuration at some chiral centers. Sucralose has total (x) chiral centers, it is (y) times as sweet as cane sugar, and its molecular mass increases by (z) g/mol on acylation with excess acetyl chloride. Calculate the value of (x + y + z).

5) A cyclic hexapeptide (molecular weight = 488) upon complete hydrolysis gives natural amino

acids glycine, alanine, phenyl alanine & valine. Glycine contributes 37.8 % to total weight of the hydrolysed products. What is the total number of alanine units in this cyclic hexapeptide ?

6) An open chain organic compound (X) consists of C, H, O only. It gives positive 2,4-DNP test and positive iodoform test. On reaction with $\text{NH}_2\text{OH}/\text{H}^+$, it gives 2 different organic compounds (Y) and (Z), which are stereoisomers of each other. When (Y) and (Z) are separately heated with conc. H_2SO_4 , we obtain two different organic compounds (A) and (B) respectively, which are structural isomers of each other. What is the number of carbon atoms in (X) having lowest possible molecular mass.

ANSWER KEYS

PART-1 : MATHEMATICS

SECTION-I (i)

Q.	1	2	3	4
A.	C	B	C	A

SECTION-I (ii)

Q.	5	6	7
A.	B,C,D	A,B,C	B,C,D

SECTION-I (iii)

Q.	8	9	10	11
A.	B	C	C	B

SECTION-II

Q.	12	13	14	15	16	17
A.	58	1	62	9	336	1

PART-2 : PHYSICS

SECTION-I (i)

Q.	18	19	20	21
A.	B	A	C	C

SECTION-I (ii)

Q.	22	23	24
A.	A,C	B,C	A,B,C

SECTION-I (iii)

Q.	25	26	27	28
A.	A	B	A	A

SECTION-II

Q.	29	30	31	32	33	34
A.	4	24	12	24	24	2

PART-3 : CHEMISTRY

SECTION-I (i)

Q.	35	36	37	38
A.	C	C	B	B

SECTION-I (ii)

Q.	39	40	41
A.	A,B	A,C,D	C

SECTION-I (iii)

Q.	42	43	44	45
A.	B	D	B	C

SECTION-II

Q.	46	47	48	49	50	51
A.	29	2	3	819	1	4

SOLUTIONS

PART-1 : MATHEMATICS

$$\begin{aligned}
 1) S &= {}^{99}C_0 - {}^{99}C_2 + {}^{99}C_4 - {}^{99}C_6 + \dots - {}^{99}C_{98} \\
 (1+x)^{99} &= {}^{99}C_0 + {}^{99}C_1 x + {}^{99}C_2 x^2 + {}^{99}C_3 x^3 + \dots + {}^{99}C_{99} x^{99} \\
 (1-x)^{99} &= {}^{99}C_0 - {}^{99}C_1 x + {}^{99}C_2 x^2 - \dots - {}^{99}C_{99} x^{99}
 \end{aligned}$$

$$\frac{(1+x)^{99} + (1-x)^{99}}{2} = {}^{99}C^0 + {}^{99}C_2 x^2 + {}^{99}C_4 x^4 + \dots + {}^{99}C_{98} x^{98}$$

Put $x = i$

$$\begin{aligned}
 S &= \frac{(1+i)^{99} + (1-i)^{99}}{2} = {}^{99}C^0 \\
 S &= \frac{2^{\frac{99}{2}} \left[\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right]^{99} + 2^{\frac{99}{2}} \left[\cos \frac{\pi}{4} - i \sin \frac{\pi}{4} \right]^{99}}{2}
 \end{aligned}$$

$$S = \frac{2^{\frac{99}{2}} \cdot 2 \cos \frac{99\pi}{4}}{2} = 2^{49.5} \cdot \left(-\frac{1}{\sqrt{2}} \right) = -2^{49}$$

$$2) \text{ Let us consider a complex no. } z \text{ such that } z^2 = \frac{2+i}{3-i} \text{ \& } n = 2(2s+1)$$

$$f(x) = x^n - 1 = (x - \alpha_0)(x - \alpha_1) \dots (x - \alpha_{n-1})$$

$$f\left(\frac{z}{i}\right) = \left(\frac{1}{i}\right)^n (z - i\alpha_0)(z - i\alpha_1)(z - i\alpha_2) \dots (z - i\alpha_{n-1})$$

$$f\left(\frac{-z}{i}\right) = \left(\frac{-1}{i}\right)^n (z + i\alpha_0)(z + i\alpha_1) \dots (z + i\alpha_{n-1})$$

$$f\left(\frac{z}{i}\right) f\left(\frac{-z}{i}\right) = (z^2 + \alpha_0^2)(z^2 + \alpha_1^2) \dots (z^2 + \alpha_{n-1}^2)$$

$$\prod_{k=0}^{n-1} [(2+i) + (3+i) \alpha_k^2]$$

$$= (3-i)^n \prod_{k=0}^{n-1} \left(\frac{2+i}{3-i} + \alpha_k^2 \right)$$

$$= (3-i)^n \prod_{k=0}^{n-1} (z^2 + \alpha_k^2)$$

$$= (3-i)^n f\left(\frac{z}{i}\right) f\left(\frac{-z}{i}\right)$$

$$= (3-i)^n \left[(z^2)^{2s+1} + 1 \right]^2$$

$$= (3-i)^n \left[\left(\frac{2+i}{3-i} \right)^{2s+1} + 1 \right]^2$$

$$= (3-i)^{2(2s+1)} \left(\frac{(2+i)^{2s+1} + (3-i)^{2s+1}}{(3-i)^{2s+1}} \right)^2$$

$$= \left[(2+i)^{n_2} + (3-i)^{n_2} \right]^2$$

3) coefficients of y^m in $e^{xy} (e^y - 1)^m$

= coefficient of y^m in

$$e^{xy} \left[e^{my} - {}^m C_1 e^{(m-1)y} + {}^m C_2 e^{(m-2)y} + \dots (-1)^m {}^m C_m \right]$$

$$= \text{coefficient of } y^m \text{ in } e^{xy} \left[\left(1 + y + \frac{y^2}{2} + \dots \right) - 1 \right]^m$$

$$= \text{coefficient of } y^m \text{ in } e^{xy} y^m \left[\left(1 + \frac{y}{2} + \frac{y^2}{2} + \dots \right) \right]^m$$

= coefficient of y^m in

= coefficient of y^m in

$$\left(e^{y(x+m)} - {}^m C_1 e^{y(x+m-1)} + {}^m C_2 e^{y(x+m-2)} \dots (-1)^m {}^m C_m e^{xy} \right)$$

$$= (x+m)^m - {}^m C_1 (x+m-1)^m + {}^m C_2 (x+m-2)^m + \dots (-1)^m {}^m C_m x^m$$

$$= \underline{m}$$

4) Let $Z = r(\cos \theta + i \sin \theta)$

$$\left| 2z + \frac{1}{z} \right| = 1$$

$$\left| \left(2r + \frac{1}{r} \right) \cos \theta + i \left(2r - \frac{1}{r} \right) \sin \theta \right| = 1$$

$$\left(2r + \frac{1}{r} \right)^2 \cos^2 \theta + \left(2r - \frac{1}{r} \right)^2 \sin^2 \theta = 1$$

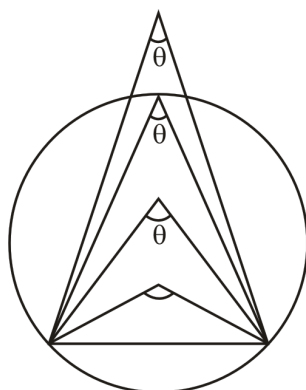
$$\Rightarrow 4r^2 + \frac{1}{r^2} + 3 = 8\sin^2 \theta$$

$$\text{Now } \frac{4r^2 + \frac{1}{r^2}}{2} \geq 2 \text{ (use AM} \geq \text{GM)}$$

$$\Rightarrow 4r^2 + \frac{1}{r^2} \geq 4 \Rightarrow 8\sin^2 \theta \geq 7$$

$$\Rightarrow \sin^2 \theta \geq \frac{7}{8} = 0.87$$

5)



(A) If $\arg\left(\frac{z_1}{z_2}\right) = \frac{\pi}{2}$ then z_1 and z_2

Subtend right angle at circum centre origin.

□ The chord joining z_1 and z_2 will subtend an angle θ at z such that

$$\theta = \frac{\pi}{4} \text{ if } |z| = 1$$

$$\theta < \frac{\pi}{4} \text{ if } |z| > 1, \theta > \frac{\pi}{4} \text{ if } |z| < 1$$

$$\begin{aligned} \text{(B)} \quad & |z_1 z_2 + z_2 z_3 + z_3 z_1| \\ &= |z_1| |z_2| |z_3| \left| \frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3} \right| \\ &= \frac{|z_1 + z_2 + z_3|}{\left(\frac{(z_1 + z_2)(z_2 + z_3)(z_3 + z_1)}{z_1 z_2 z_3} \right)} \end{aligned}$$

$$\begin{aligned} \text{(C)} \quad &= \frac{(\bar{z}_1 + \bar{z}_2)(\bar{z}_2 + \bar{z}_3)(\bar{z}_3 + \bar{z}_1)}{\bar{z}_1 \bar{z}_2 \bar{z}_3} \\ &= \frac{(z_1 + z_2)(z_2 + z_3)(z_3 + z_1)}{z_1 z_2 z_3} \end{aligned}$$

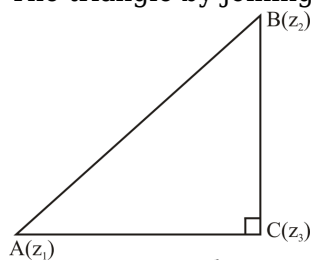
□ If $z = \bar{z} \Rightarrow \text{Im}(z) = 0$

$$\text{Im} \left(\frac{(z_1 + z_2)(z_2 + z_3)(z_3 + z_1)}{z_1 z_2 z_3} \right) = 0$$

(D) Let $A(z_1)$, $B(z_2)$ and $C(z_3)$

$$BC = CA \text{ and } AB = \sqrt{2}BC$$

The triangle by joining z_1 , z_3 and z_2 is isosceles and right angled at z_3 .



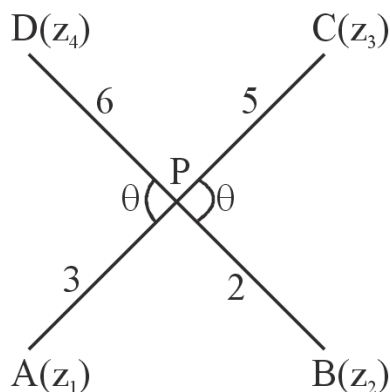
$$\begin{aligned} \frac{z_3 - z_1}{z_3 - z_2} &= \frac{|z_3 - z_1|}{|z_3 - z_2|} e^{\pm i\pi/2} \\ \Rightarrow \frac{z_3 - z_1}{z_3 - z_2} &= \pm i \Rightarrow \text{Re} \left(\frac{z_3 - z_1}{z_3 - z_2} \right) = 0 \end{aligned}$$

Let A, B, C, D denote the complex numbers z_1, z_2, z_3 & z_4 respectively in the argand plane.

$$\text{Now } \frac{5z_1 + 3z_3}{8} = \frac{6z_2 + 2z_4}{8}$$

P divides AC in the ratio 3 : 5

& P divides BD in the ratio 1 : 3.



Let $AP = 3\lambda$, $CP = 5\lambda$

$BP = k$, $DP = 3k$

then in $\triangle APD$ $AD^2 = AP^2 + DP^2 - 2(AP)(DP)\cos\theta$

$$\Rightarrow |z_1 - z_4|^2 = 9\lambda^2 + 9k^2 - 2(3\lambda)(3k)\cos\theta$$

$$\text{in } \triangle PBC \Rightarrow |z_2 - z_3|^2 = 25\lambda^2 + k^2 - 2(5\lambda)(k)\cos\theta$$

$$\text{Given } 5|z_1 - z_4|^2 = 9|z_2 - z_3|^2$$

$$\Rightarrow 5 \times 9(\lambda^2 + k^2) = 9(25\lambda^2 + k^2) \Rightarrow k^2 = 5\lambda^2$$

$\therefore AP \times PC = BP \times PD \Rightarrow A, B, C, D$ are concyclic.

Final Answer : Option (2), (3) and (4)

8) (P) Total number of three digit numbers = $9 \times 10 \times 10$.

Half of which will have sum of digit even

$$(Q) \quad xyz = 140 = 2^2 \cdot 5 \cdot 7$$

Therefore number of positive integral solutions = $3 \times 3 \times {}^4C_2 = 54$

(R) $x + y + z + t = 10$, where $1 \leq x, y, z \leq 8$, $0 \leq t \leq 7$

Therefore desired number of solutions = coeff. of x^{10} in

$$(x + x^2 + \dots + x^8)^3 \times (1 + x + x^2 + \dots + x^7) = {}^{10}C_7 = {}^{10}C_7 = {}^{10}C_3 = 120$$

(S) We must have $i^3 + ai^2 + bi + c = 0$ and $(-i)^3 + a(-i)^2 + b(-i) + c = 0$
 $\Rightarrow b = 1$ and $a = c$

Therefore number of numbers of type abc or cba is ${}^9C_1 = 9$

Number of number of type bac or bca is ${}^{10}C_1 = 10$

But 111 is included in both the counting.

$$9) \quad z_k = \cos\left(\frac{2k\pi}{10}\right) + i \sin\left(\frac{2k\pi}{10}\right), \quad k = 1, 2, \dots, 9$$

$$z_k = e^{i\left(\frac{2k\pi}{10}\right)}$$

It is easy to see that z_k are 10^{th} roots of unity except $z = 1$.

$$(P) \quad z_k \cdot z_j = 1$$

$$\Rightarrow z_j = \frac{1}{z_k} = e^{-i(2k/10)} = z_{10-k}$$

True

$$(Q) \quad z_i \cdot z = z_k$$

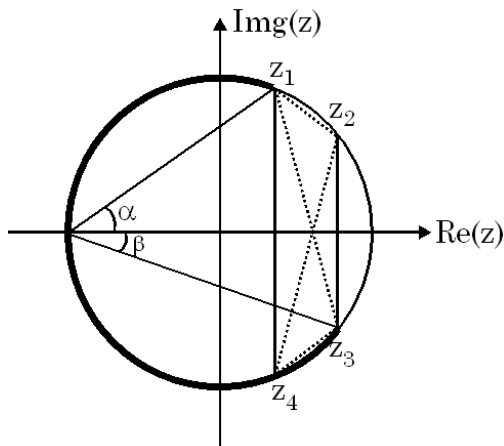
$$\Rightarrow e^{i\frac{2\pi}{10}} \cdot z = e^{i\left(\frac{2k\pi}{10}\right)}$$

False

$$\begin{aligned}
 \text{(R)} \quad (z^{10} - 1) &= (z - 1)(z^9 + z^8 + z^7 + \dots + 1) \\
 (z - 1)(z - z_1)(z - z_2) \dots (z - z_9) &= (z - 1)(z^9 + z^8 + \dots + 1) \\
 \Rightarrow (z - z_1)(z - z_2) \dots (z - z_9) &= (z^9 + z^8 + \dots + 1) \\
 \text{Put } z = 1 \text{ and take modulus} \\
 \frac{|(1 - z_1)(1 - z_2) \dots (1 - z_9)|}{10} &= 1
 \end{aligned}$$

$$\begin{aligned}
 \text{(S)} \quad 1 - \sum_{k=1}^9 \cos\left(\frac{2k\pi}{10}\right) \\
 = 1 - \left[\cos \frac{2\pi}{10} + \cos \frac{4\pi}{10} + \dots + \cos \frac{18\pi}{10} \right] \\
 = 1 - (-1) = 2
 \end{aligned}$$

$$\begin{aligned}
 10) \quad z^4 - 6z^3 + 18z^2 - 30z + 25 = 0 \quad z_1 = 1 + 2i \quad \therefore z_4 = 1 - 2i \\
 \text{let } z_1 = \alpha + i\beta, z_3 = \alpha - i\beta \\
 z_1 + z_2 + z_3 + z_4 = 6 \Rightarrow 2 + 2\alpha = 6 \Rightarrow \alpha = 2 \\
 \& \quad z_1 z_2 z_3 z_4 = 25 \Rightarrow (1+4)(4+\beta^2) = 25 \Rightarrow \beta = \pm 1 \\
 \Rightarrow z_2 = 2 + i : z_3 = 2 - i
 \end{aligned}$$



$$\text{(P) Area bounded by } z_1 z_2 z_3 z_4 = \frac{1}{2}(6)(1) = 3$$

$$\text{(Q) } \frac{|z_0 - \bar{z}_1|}{|z_0 - \bar{z}_4|} \cdot \frac{|z_0 - \bar{z}_3|}{|z_0 - \bar{z}_2|} = 1$$

$$\text{(R) } z \text{ is diagonal intersection of } z_1 z_3 \text{ and } z_2 z_4$$

$$\therefore z = \frac{5}{3}$$

$$\text{(S) } \alpha + \beta = \tan^{-1}\left(\frac{2}{1+\sqrt{5}}\right) + \tan^{-1}\left(\frac{1}{2+\sqrt{5}}\right)$$

z lies on major arc of circle containing $z_1 z_3$ & centred at origin

$$\frac{1}{\sqrt{5}}|z - 2 - i|_{\max} = \frac{1}{\sqrt{5}} \times 2\sqrt{5} = 2$$

$$\begin{aligned}
 11) \quad P(n+1) &= \sum_{k=0}^n (-1)^{n-k} \left\{ 1 - \frac{1}{k+1} \right\} \frac{\frac{n+1}{k}}{\frac{n+1-k}{k}} \\
 &= \sum_{k=0}^n (-1)^{n-k} {}^{n+1}C_k - \sum_{k=0}^n (-1)^{n-k} \frac{\frac{n+1}{k+1} (n+2)}{\frac{n+1-k}{k+1}} \cdot \frac{1}{n+2}
 \end{aligned}$$

$$\begin{aligned}
&= 1 - \frac{1}{n+2} \sum_{k=0}^n (-1)^{n-k} {}^{n+2}C_{n+1-k} \\
&= 1 - \frac{1 + (-1)^n}{n+2}
\end{aligned}$$

12) **Solution :**

$$\begin{aligned}
(1+x)^{\frac{1}{x}} &= e^{\log_e (1+x)^{\frac{1}{x}}} = e^{\frac{1}{x} \log_e (1+x)} \\
&= e^{\frac{1}{x} \left(x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots \infty \right)} \\
&= e^{1 - \frac{x}{2} + \frac{x^2}{3} - \frac{x^3}{4} + \dots \infty} \\
&= e \cdot e^{-\frac{x}{2} + \frac{x^2}{3} - \frac{x^3}{4} + \dots \infty} \\
&= e \left[1 + \left(-\frac{x}{2} + \frac{x^2}{3} - \frac{x^3}{4} + \dots \infty \right) + \frac{1}{2!} \left(-\frac{x}{2} + \frac{x^2}{3} - \frac{x^3}{4} + \dots \infty \right)^2 + \frac{1}{3!} \left(-\frac{x}{2} + \frac{x^2}{3} - \frac{x^3}{4} + \dots \infty \right)^3 + \dots \infty \right] \\
&= e \left[1 - \frac{x}{2} + \left(\frac{1}{3} + \frac{1}{2!} \cdot \frac{1}{2^2} \right) x^2 + \left\{ \frac{-1}{4} + \frac{1}{2!} \cdot 2 \cdot \frac{-1}{2} \cdot \frac{1}{3} + \frac{1}{3!} \cdot \left(\frac{-1}{2} \right)^3 \right\} x^3 + \dots \right] \\
&= e \left(1 - \frac{x}{2} + \frac{11}{24} x^2 - \frac{7}{16} x^3 + \dots \infty \right)
\end{aligned}$$

Given by $a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \dots \infty$

$$\square \quad a_2 = \frac{11e}{24} = \frac{m}{n} \cdot e \quad \Rightarrow m = 11, n = 24$$

$$\text{and } a_3 = \frac{-7e}{16} = \frac{p}{q} \cdot e \quad \Rightarrow p = -7 \text{ \& } q = 16$$

$$\square \quad |m| + |p| + n + q = 11 + 7 + 24 + 16 = 58$$

Final Answer : 58

13) **Solution :**

Using the identity

$$a^3 + b^3 + c^3 - 3abc = (a + b + c)(a + b\omega + c\omega^2)(a + b\omega^2 + c\omega)$$

where ω is non-real cube root of unity, we have

$$\begin{aligned}
a + b + c &= \sum_{n=0}^{\infty} \frac{x^{3n}}{(3n)!} + \frac{x^{3n+1}}{(3n+1)!} + \frac{x^{3n+2}}{(3n+2)!} \\
\sum_{m=0}^{\infty} \frac{x^m}{m!} &= e^x \\
a + b\omega + c\omega^2 &= \sum_{n=0}^{\infty} \frac{x^{3n}}{(3n)!} + \frac{(\omega x)^{3n+1}}{(3n+1)!} + \frac{(\omega x)^{3n+2}}{(3n+2)!} \\
&= \sum_{n=0}^{\infty} \frac{(\omega x)^{3n}}{(3n)!} + \frac{(\omega x)^{3n+1}}{(3n+1)!} + \frac{(\omega x)^{3n+2}}{(3n+2)!} \\
&= \sum_{m=0}^{\infty} \frac{(\omega x)^m}{m!} = e^{\omega x}
\end{aligned}$$

$$\text{and } a + b\omega^2 + c\omega = \sum_{n=0}^{\infty} \frac{x^{3n}}{(3n)!} + \frac{\omega^2 (x^{3n+1})}{(3n+1)!} + \frac{\omega (x^{3n+2})}{(3n+2)!}$$

$$\begin{aligned} & \{\text{writing } \omega^2 = (\omega^2)^{3n+1} \text{ \& } \omega = (\omega^2)^{3n+2}\} \\ &= \sum_{n=0}^{\infty} \frac{x^{3n}}{(3n)!} + \frac{(\omega^2 x)^{3n+1}}{(3n+1)!} + \frac{(\omega^2 x)^{3n+2}}{(3n+2)!} \\ &= \sum_{m=0}^{\infty} \frac{(\omega^2 x)^m}{m!} = e^{\omega^2 x} \end{aligned}$$

$$\square a^3 + b^3 + c^3 - 3abc = (a + b + c)(a + b\omega + c\omega^2)(a + b\omega^2 + c\omega)$$

$$= e^x \cdot e^{\omega x} \cdot e^{\omega^2 x} = e^{(1+\omega+\omega^2)x}$$

$$= e^{0 \cdot x} = e^0 = 1$$

{Using $1 + \omega + \omega^2 = 0$, where ω is the non-real cube root of unity.}

Final Answer : 1

$$14) \sum_{r=0}^{2014} \sum_{k=0}^r (-1)^k (k+1)(k+2) {}^{2019}C_{r-k} = \sum_{r=0}^{2014} 2 {}^{2016}C_r = 2^{2017} - 4034$$

$$15) \boxed{} \boxed{} \boxed{} \boxed{} \boxed{}$$

1st blank can be filled in 9 ways

2nd blank can be filled in 9 ways (repetition is allowed)

3rd blank can be filled in 9 ways

4th blank can be filled in 9 ways

Now we have to fill the 5th blank carefully such that the number is divisible by 3

Add the 4 numbers in the first 4 blanks .

If their sum is in the form $3n$, then fill the last blank by 3, 6 or 9

If their sum is in the form $3n + 1$, then fill the last blank by 2, 5 or 8

If their sum is in the form $3n + 2$, then fill the last blank by 1, 4 or 7

So last blank can be filled by 3 ways only

$$\square \text{ Required numbers} = 9 \times 9 \times 9 \times 9 \times 3 = 3^9$$

16)

Question Explanation: Circular arrangement with separation constraints.

Concept: Principle of Inclusion-Exclusion (PIE).

Solution: Total people = 7. We need to find the number of ways to arrange them around a circular table such that no two Americans (A1, A2) are together and no two British (B1, B2) are together. We use PIE: Total circular arrangements minus arrangements where at least one pair of same nationality sits together.

- Total arrangements: $(7-1)! = 6! = 720$

- Case X (A1, A2 together): Treat (A1A2) as one unit. $(6-1)! \times 2! = 120 \times 2 = 240$

- Case Y (B1, B2 together): Similarly, $(6-1)! \times 2! = 240$

- Case $X \cap Y$ (Both pairs together): Treat (A1A2) and (B1B2) as units. $(5-1)! \times 2! \times 2! = 24 \times 4 = 96$

Favorable ways: Total $-(X+Y) + (X \cap Y)$
 $= 720 - (240 + 240) + 96 = 720 - 480 + 96 = 336$

Final Answer: Option (3)

$$\begin{aligned}
& 17) \quad e^{i2m \cot^{-1} x} \left\{ \frac{xi+1}{xi-1} \right\}^m \\
& \left\{ e^{i \cot^{-1} x} \left(\frac{xi+1}{xi-1} \right) \right\}^m \\
& \text{Take } \cot^{-1} x = \theta \\
& x = \cot \theta \\
& \tan \theta = 1/x \\
& \left\{ (\cos 2\theta + i \sin 2\theta) \left(\frac{xi+1}{xi-1} \right) \right\}^m \\
& \Rightarrow \left\{ \left(\frac{1 - \frac{1}{x^2}}{1 + \frac{1}{x^2}} + \frac{i \frac{2}{x}}{1 + \frac{1}{x^2}} \right) \left(\frac{xi+1}{xi-1} \right) \right\}^m \\
& \Rightarrow \left\{ \left(\frac{x^2 - 1 + 2ix}{x^2 + 1} \right) \left(\frac{xi+1}{xi-1} \right) \right\}^m \\
& \Rightarrow \left\{ \frac{x^2 - 1 + 2ix}{x^2 + 1} \cdot \frac{(ix+1)(ix-1)}{(ix-1)^2} \right\}^m \\
& \Rightarrow \left\{ \frac{x^2 - 1 + 2ix}{x^2 + 1} \cdot \frac{-x^2 - 1}{-x^2 + 1 - 2ix} \right\}^m \\
& \Rightarrow (1)^m = (1)
\end{aligned}$$

PART-2 : PHYSICS

18)

Correct Answer is (B)

19)

Correct Answer is (A)

20)

Correct Answer is (C)

21)

Correct Answer is (C)

22)

Correct Answer is (A,C)

23)

Correct Answer is (B, C)

24)

Correct Answer is (A,B,C)

25)

Correct Answer is (A)

26)

Correct Answer is (B)

27)

Correct Answer is (A)

28)

Correct Answer is (A)

29)

Correct Answer is (4)

30)

Correct Answer is (24)

31)

Correct Answer is (12)

32)

Correct Answer is (24)

33)

Correct Answer is (24)

34)

Correct Answer is (2)

PART-3 : CHEMISTRY

35) Double Helix Structure

36)

Thermoplastic polymers do not possess extensive cross linking by covalent bonds.

37) (A) it is a hydrogenated polymer of isoprene

(B) alternate copolymer

(C) cannot be vulcanized

(D) Monomers are unsaturated

38)

Basic amino acids are positively charged at $\text{pH} = 7$.

39)

The rate of reaction is dependent on the concentration of cyanide ion.

The rate of reaction increases if Cl is replaced by F.

40) (A) Cationic detergents are Quaternary ammonium salts of amines with acetates, chlorides (or) bromides as anions. These detergents have germicidal properties

(B) Bacteria cannot degrade the detergents containing highly branched chains, therefore, in most of the detergents used these days, the branching is kept to a minimum so that the detergents become easily biodegradable

(C) Some synthetic detergents can give foam even in ice cold water.

(D) Synthetic detergents are cleansing agents which have all the properties of soaps, but which actually do not contain any soap

41)

It is vitamin-C

42)

P - 1, 3; Q - 4; R - 2, 3; S - 5

43) The correct answer is option (D)

44)

The correct answer is option (B)

45)

The correct answer is (C)

46)

A = 9, B = 10, C = 10 (Haworth synthesis)

47) The two monomers are sebacic acid and hexamethylene diamine (lower mass). Two moles N_2 are evolved from two moles monomer.

48)

The disaccharide has acetal linkage. The reaction occurs at anomeric carbons of both monosaccharides. (1 + 2)

49) (x) = 9, (y) = 600, (z) = $5 \times 42 = 210$

50)

It is cyclic therefore has 6 peptide bonds. It contains 3 units of glycine and one unit each of alanine, phenyl alanine & valine.

$$488 + 6 \times 18 = 3 \times 75 + 89 + 165 + 117$$

51) (X) is butanone. CH_3CHO gives abnormal Beckmann so it can not be (X)